

A Two-Stage Conditional Diffusion Model With Differential Attention for Hyperspectral and Multispectral Image Fusion

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Abstract—Diffusion models have been used extensively for hyperspectral and multispectral image fusion; however, their intrinsic hallucination phenomenon frequently results in a loss of high-frequency details in the fused images. To address this limitation, this paper proposes a fusion method that integrates a differential attention mechanism with a two-stage conditional diffusion model. The proposed method leverages differential attention to compute the difference between two independent attention maps, thereby producing sparser and more focused attention representations. In addition, the two-stage conditional injection strategy is implemented to realize precise control over the image generation process. In the first stage, feature-level linear modulation via affine transformation is applied within the encoder to maintain global structural consistency. Then, in the second stage, wavelet features extracted from the conditioning images are injected into the decoder to facilitate the restoration of fine-grained details. Extensive validation experiments on the CAVE, Harvard, Pavia Center and Chikusei datasets verify the effectiveness of the proposed method. Compared with numerous state-of-the-art approaches, our method consistently achieves superior performance across key evaluation metrics. On the CAVE dataset, the ERGAS and RMSE metrics improved by 2.16% and 5.01%, and these metrics increased by 1.78% and 1.38% on the Pavia Center dataset, respectively. The code will be available at <https://github.com/Ruijie2580/DifferentialDiff>

Index Terms—Diffusion model, hyperspectral, multispectral, image fusion, differential attention.

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I. INTRODUCTION

RECENTLY, deep learning-based hyperspectral image fusion methods have emerged as the predominant paradigm for HSI-MSI fusion. Early fusion approaches primarily relied upon convolutional neural networks (CNNs) [1], [2]; nevertheless, the inherent locality of standard convolutional operations restricts their capacity to model long-range spatial-spectral dependencies. To overcome this limitation, subsequent approaches attempted to expand the effective receptive field. For example, the MSDCNN [3] and BDPN [4] methods leverage multiscale convolution and multiresolution analysis, respectively. However, since their design philosophies partially inherit from conventional component substitution frameworks, these models may introduce an imbalance between spatial detail enhancement and spectral fidelity. To address this issue, Qu et al. [5] proposed a densely connected network architecture to promote a more effective flow and feature integration across layers, thereby mitigating spectral and spatial distortions. As the field matured, a parallel line of research focused on improving model efficiency and generalizability. To address the substantial computational overhead and diminished generalizability frequently associated with deeper networks, Ran et al. [6] engineered a dual-branch architecture that utilizes high-resolution features as a guiding prior, and they realized a significant reduction in model parameters while boosting generalizability. Similarly, CSLP [7] integrates compressed sensing [8] with attention mechanisms to capture global contexts more efficiently. A separate, yet equally critical, challenge is the limited interpretability of many deep learning models. To enhance model transparency, DPDU-Net [9] employs a deep unfolding network [10], [11], [12] that integrates both implicit and semi-implicit priors, explicitly linking the network architecture to a conventional optimization algorithm. In addition, SINet [13] integrates the long-range sequence modeling capability of Mamba [14] with convolutional networks. By combining multi-scale reversible spatial modules with Haar wavelet decomposition, it enhances spatial feature discrimination and achieves a favorable balance between detail preservation and spectral fidelity through a spectral compression strategy.

Despite the significant progress achieved in these previous studies, the inductive bias inherent to CNNs imposes a fundamental constraint on their ability to model long-range dependencies [15], and this performance bottleneck has

motivated a paradigm shift toward architectures centered on the Transformer [16] and its global self-attention mechanism. Foundational vision Transformers, e.g., the Swin Transformer [17] have demonstrated strong feature extraction capabilities; however, their generic architectures were not optimized for the unique characteristics of remote sensing data. Consequently, a series of specialized models was developed. For example, PSRT [18] reengineered the shuffling strategy of the Shuffle Transformer [19] to model global information exchange efficiently, leveraging the strong correlation and complementarity between remote sensing image patches. However, its design did not explicitly account for the spectral dimension. To process spatial and spectral information concurrently, SST [20] adopted a dual-branch Transformer architecture, and a critical observation emerged from these efforts, i.e., despite their proficiency in capturing global context, Transformer-based models exhibit an intrinsic limitation in terms of reconstructing high-frequency details. Subsequent research has attempted to remedy this deficiency. Wang et al. proposed PMA²-Net [21], which employs multi-scale axial attention and progressive fusion to strengthen spectral-spatial feature interactions while preserving both spatial details and spectral information. MIMOSST [22] introduced multiscale supervision during training to balance global and local feature reconstruction, and HCMSSST [23] engineered intricate cross-layer feature interaction pathways to preserve finer details. These elaborate designs partially alleviate detail blurring; however, they primarily function as post hoc corrections rather than fundamental solutions. SSDAN [24] employs a spatial-frequency dual-branch architecture with differential complementary modules to alleviate feature mismatch-induced distortions. Moreover, recent studies have explored various structural priors and collaborative mechanisms, including structure self-representation [25], spatial-spectral structure preservation [26], cross-view consistency and matching [27], cross-modal semantic collaboration [28], and multi-domain adaptation [29]. These mechanisms have achieved remarkable success in clustering, segmentation, target detection, and multi-source classification tasks, highlighting the importance of preserving intrinsic structures and promoting semantic consistency across heterogeneous modalities. More recently, the rapid development of multi-modal representation learning and remote sensing foundation models has further emphasized unified feature modeling and cross-modal knowledge transfer [30], [31], providing valuable insights for structural information preservation and heterogeneous feature interaction in HSI-MSI fusion.

These studies highlight the importance of frequency-domain representations, structural priors, and heterogeneous feature interactions for preserving fine-grained details. Nevertheless, accurately reconstructing high-frequency textures remains challenging for existing methods. In this context, diffusion models [32], [33], which perform image generation through iterative denoising, provide a promising framework for progressive detail refinement and high-fidelity reconstruction.

A diffusion model operates on a two-step principle. First, a forward process gradually introduces Gaussian noise to an image until it becomes pure noise. Second, a learnable reverse process works in reverse, iteratively removing the

noise to reconstruct the original high-fidelity image. This iterative refinement process enables the model to progressively reconstruct image details across multiple scales, demonstrating exceptional performance in the restoration of high-frequency textures. As a result, diffusion models have recently been applied in the image fusion field [34], exhibiting considerable potential. Initial efforts, e.g., PanDiff [35], adapted the diffusion framework from super-resolution to the fusion task. Building on this, CrossDiff [36] adopted a dual-branch architecture to facilitate direct interaction between MSI and HSI information in each step of the generation process. In addition, to enforce more robust cross-modality constraints, S2CycleDiff [37] incorporated the cycle consistency principle, constructing a bidirectional “spatial-spectral” diffusion loop. Other studies have investigated hybrid strategies. For example, the SSDiff [38] method integrates subspace decomposition [39] and a low-rank adaptation [40] fine-tuning process to enhance fusion quality. In a notable effort to bridge deep learning with conventional techniques, the PLRDiff [41] method leverages pretrained diffusion models as a Bayesian prior within a model-driven optimization framework, significantly improving model interpretability. To address the issue of naive condition injection in existing diffusion models, PanDiT [42] proposes a Diffusion Transformer [43] based framework that achieves efficient and high-fidelity pansharpening in only two inference steps by decoupling the injection of spatial and time-frequency features and integrating implicit sampling.

Existing diffusion-based methods have improved fusion quality; however, they remain susceptible to a critical issue whereby the generative process fabricates textures that, while visually plausible, lack fidelity to the source imagery [44]. The generation of such artifacts fundamentally compromises the reliability of the fusion outcomes. To address this problem, we propose a fusion method built around a differential attention diffusion model. The core innovation of this model is a differential attention mechanism that generates two distinct attention maps simultaneously and calculates the difference between them, which effectively filters noise and suppresses the activation of irrelevant features, thereby producing a sparser, more focused attention map. This mechanism compels the model to focus on regions of significant semantic divergence between the source images, which is critical in terms of realizing accurate modeling and reconstructing key details. In addition, to establish precise control over the generation process, we design a two-stage conditional injection strategy that incorporates a feature-level linear modulation layer (implemented via affine transformation) to inject global, coarse-grained conditional information, which is complemented by a wavelet feature injection module that explicitly operates in the frequency domain to guide the fine-grained restoration of both texture and structure.

The primary contributions of this study are summarized as follows:

- A conditional diffusion framework for hyperspectral and multispectral image fusion is proposed. By introducing degraded image information as conditional guidance, the reverse denoising process is effectively constrained,

TABLE I
LIST OF MAIN NOTATIONS USED IN THIS PAPER

Notations	Explanation
t	The timestep
T	Maximum number of diffusion steps in the diffusion model
x_t	An image at timestep t
ϵ	Gaussian noise
c	The Diffusion condition
$\mathcal{X}$	The LRHS image
$\mathcal{Y}$	The HRMS image
$\mathcal{Z}$	The original HRHS image
$\hat{\mathcal{Z}}$	The fusion image
LL, LH, HL, HH	The wavelet coefficients
$[x_{\tau_0}, \dots, x_{\tau_S}]$	DDIM sampling sequence, S is the number of sampling steps
F_l	Model feature map of layer l

mitigating diffusion-induced hallucinations and improving both spectral and spatial fidelity.

- A differential attention mechanism is introduced to enhance selective modeling of discriminative spectral-spatial information. By emphasizing structural differences and suppressing redundant responses, the proposed mechanism improves fine-grained detail representation and reconstruction stability.
- A two-stage conditional injection strategy is designed to provide hierarchical spatial-spectral guidance. Specifically, the FiLM module modulates global structural and spectral information, while the WCA module introduces frequency-domain detail constraints, enabling coarse-to-fine reconstruction while improving spectral consistency, spatial clarity, and structural integrity.

II. METHODOLOGY

Here, the proposed two-stage guided diffusion model, which incorporates a differential attention mechanism, is described in detail. First, we establish the fundamental notations used in the paper. We then present the overall architecture of the model, elucidating its core design principles. Finally, we present an in-depth analysis of each constituent module.

A. Notations

The notations used in this paper are defined in Table I. Specifically, we formally denote the input LRHS image and HRMS image as $\mathcal{X} \in \mathbb{R}^{C \times h \times w}$ and $\mathcal{Y} \in \mathbb{R}^{c \times H \times W}$, respectively. The output image generated by the fusion algorithm is represented as $\hat{\mathcal{Z}} \in \mathbb{R}^{C \times H \times W}$, and its corresponding true HRHS image is denoted $\mathcal{Z} \in \mathbb{R}^{C \times H \times W}$. Here, c, h, w and C, H, W represent the number of channels, height, and width of the images, where lowercase and uppercase refer to low resolution and high resolution, respectively.

B. Overall Architecture

The overall architecture of the proposed network is shown in Fig. 1. Fig. 1(b) illustrates the backbone network of the diffusion model. The network adopts a UNet-based structure comprising multiple encoder-decoder layers and two intermediate layers. In this design, information is exchanged between the corresponding encoder and decoder layers via cross-layer skip connections. Initially, the input image undergoes feature

extraction through a starting convolutional layer before being passed to the encoder module. Each encoder comprises a feature-wise linear modulation (FiLM) layer, a residual block (ResBlock), and a multi-head differential attention (MHDA) module connected in series. During the encoding stage, the LRHSI and HRMSI are concatenated along the channel dimension before being passed to the FiLM module. This process facilitates the coarse-grained injection of low-frequency features. Following deep semantic feature extraction by the encoder and intermediate layers, the resulting feature maps are passed to the decoder. Unlike the encoder, the decoder replaces the FiLM module with a wavelet cross-attention (WCA) module. In the decoding stage, the LRHS and HRMS images are first decomposed into multifrequency components using a DB1 wavelet transform, and then the WCA module performs fine-grained injection of high-frequency detail information. Finally, the decoder outputs are processed by a final convolutional layer to reconstruct the predicted image $\hat{x}_0$. Fig. 1(a) illustrates the inference process of the proposed fusion algorithm. The fusion results are generated using the DDIM sampling strategy with 25 steps.

C. Diffusion Model

A DDPM functions through two main stages, i.e. a forward diffusion process and a reverse denoising process. In the forward process, a fixed Markov chain gradually adds Gaussian noise to the input data over a series of steps. This action converts a complex data distribution into a simple distribution, e.g. a standard Gaussian distribution. Then, the reverse process, which is a learnable Markov chain, is trained to reverse this transformation. It begins with a sample from the simple distribution and iteratively removes noise to reconstruct a new data sample that is similar to the original. To achieve targeted control over the generated content, Conditional Diffusion Models are employed to incorporate external conditional information c (i.e., LRHS and HRMS images) into the reverse denoising Markov chain. Distinct from unconditional models that merely estimate the marginal distribution of the data, the proposed method learns the conditional probability distribution $p_\theta(x_{t-1}|x_t, c)$. By utilizing conditional signals to guide the denoising trajectory at each step, enabling precise attribute constraints while simultaneously recovering the true data distribution.

1) *Feature-Wise Linear Modulation (FiLM)*: To provide effective global guidance during the iterative denoising process, a FiLM [45] module is employed to inject condition-dependent modulation parameters into the encoder features. This enables the model to continuously incorporate conditional information and guides the denoising trajectory toward the target distribution. The detailed structure of the FiLM module is illustrated in Fig. 2(a).

Specifically, the guidance information comprises the LRHSI and HRMSI concatenated along the channel dimension. Subsequently, the concatenated tensor is fed into the FiLM layer. As shown in Eq. (1), the resulting output is processed by the efficient channel attention (ECA) module

$$F_{\text{ECA}} = f_{\text{conv}}^{(2C)}([\mathcal{X}, \mathcal{Y}]_{\text{I}}), \quad (1)$$

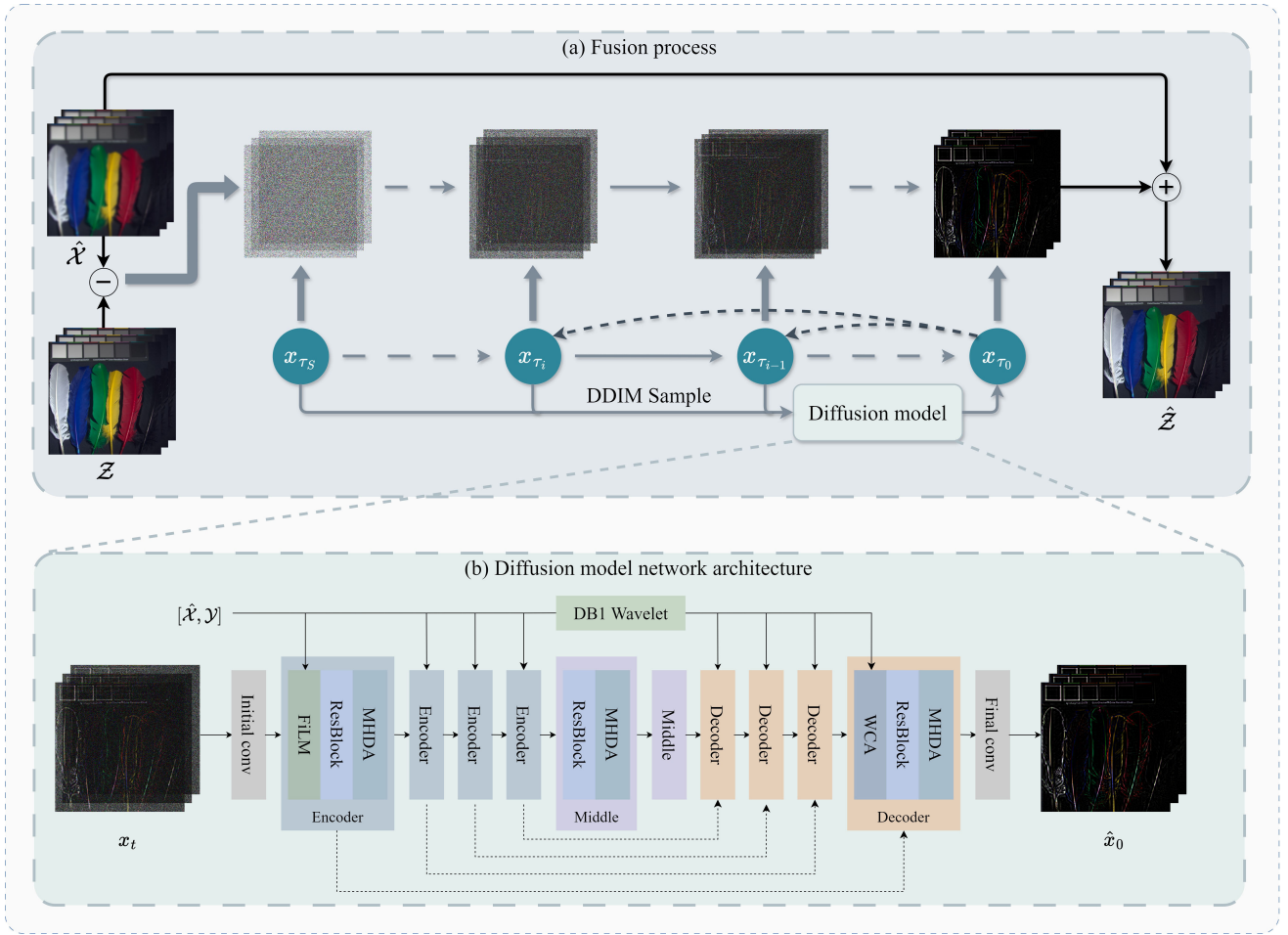


Fig. 1. The proposed model architecture is outlined. The diffusion model is designed to learn the residual between a high-resolution hyperspectral target and an upsampled low-resolution hyperspectral image. Subsequently, a DDIM sampler produces this residual image, which is then added to the upsampled low-resolution image to reconstruct the final fused result.

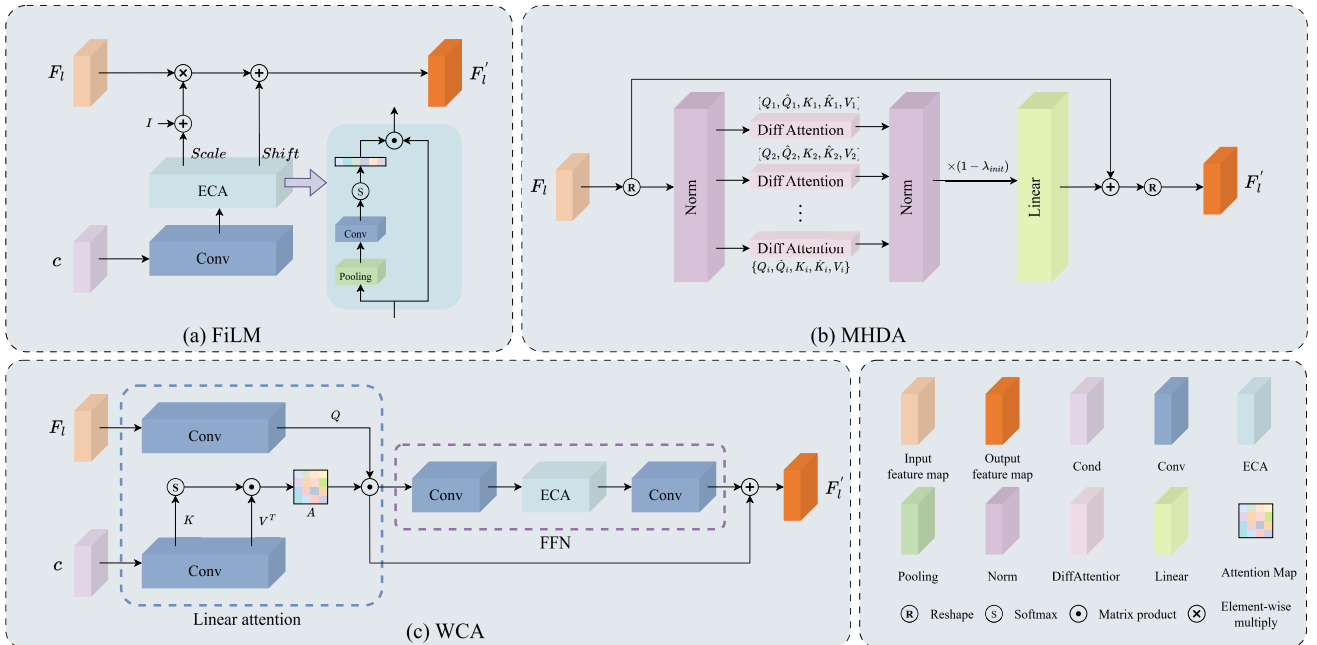


Fig. 2. Detailed architecture diagrams for each module: (a) FiLM module, (b) MHDA module, (c) WCA module.

where $f_{\text{conv}}^{(2C)}$ denotes a 3×3 convolution with $2C$ output channels, and C is the channel number of the backbone feature map. The doubled channel dimension allows the output features to be evenly split into two groups, which are used as the scale and shift parameters required by the subsequent FiLM modulation. To adaptively emphasize informative channels, the resulting features are further processed by the efficient channel attention (ECA) module, which is a lightweight channel attention mechanism composed of pooling and convolution operations. The ECA output is given by

$$F'_{\text{ECA}} = \text{softmax}(f_{\text{conv}}(f_{\text{pool}}(F_{\text{ECA}}))), \quad (2)$$

where f_{pool} denotes a pooling layer. As expressed in Eq. (3), the ECA output is split into two parts along the channel dimension to obtain the scaling factor Scale and the offset factor Shift. Subsequently, according to Eq. (4), an affine transformation is applied to the feature map F_I for coarse-grained feature injection, which can be formulated as

$$\text{Scale, Shift} = f_{\text{chunk}}(F'_{\text{ECA}}), \quad (3)$$

$$F'_I = F_I \times (\text{Scale} + I) + \text{Shift}, \quad (4)$$

where F_I , F'_I , and I denote the input feature map of the module, the output feature map of the module, and the identity matrix, respectively.

Note that the primary strength of the FiLM layer is its ability to impose global semantic constraints during the initial feature extraction stage, which ensures that the learned feature representations remain aligned with the guiding information from the outset. This early, macro-level guidance establishes a solid foundation for the decoder to subsequently perform high-fidelity detail reconstruction, which enhances the performance and stability of the entire generative model.

2) *Multi-Head Differential Attention (MHDA)*: Traditional self-attention mechanisms tend to aggregate relevant features indiscriminately, which may amplify noisy responses and lead to the accumulation of spurious structures during iterative denoising. To alleviate this issue, MHDA is introduced following the differential attention mechanism in [46]. By suppressing shared attention responses while preserving discriminative spectral-spatial correlations, MHDA mitigates attention-induced hallucinations and improves the fidelity of reconstructed hyperspectral images.

Rather than relying on a single Softmax attention map, differential attention constructs two independent sets of queries and keys, denoted as $(\mathbf{Q}, \mathbf{K})$ and $(\hat{\mathbf{Q}}, \hat{\mathbf{K}})$, to compute their respective attention responses. The final output is obtained by taking the weighted difference between the two attention maps

$$\mathbf{A} = \text{Attention}(\mathbf{Q}, \mathbf{K}), \quad (5)$$

$$\hat{\mathbf{A}} = \text{Attention}(\hat{\mathbf{Q}}, \hat{\mathbf{K}}), \quad (6)$$

$$\text{DiffAttn} = \mathbf{A} - \lambda \hat{\mathbf{A}}, \quad (7)$$

where $\text{Attention}(\cdot)$ represents the standard Softmax attention calculation, and λ is a learnable scalar employed to regulate the differential intensity. By employing this differential operation, the model effectively suppresses background noise interference, thereby focusing on more discriminative visual features.

Building upon this core operator, the Differential Attention Module illustrated in Fig. 2(b) is proposed. Here, the input feature map is first flattened into a one-dimensional sequence, which is then processed via a normalize operation to generate the normalized input for the MHDA module. This process can be formulated as

$$X_{\text{seq}} = f_{\text{norm}}(f_{\text{flatten}}(F_I)), \quad (8)$$

where f_{flatten} converts the two-dimensional input feature map into a one-dimensional sequence.

For the normalized input sequence $X_{\text{seq}} \in \mathbb{R}^{c \times hw}$, segmentation is performed along the channel dimension to obtain inputs for each attention head, which is followed by the differential attention computation defined in Eq. (7). The differential weight λ in Eq. (7) is computed as

$$\lambda = \exp(\lambda_q \cdot \lambda_k) - \exp(\lambda_{\hat{q}} \cdot \lambda_{\hat{k}}) + \lambda_{\text{init}}, \quad (9)$$

where $\lambda_q, \lambda_k, \lambda_{\hat{q}}, \lambda_{\hat{k}} \in \mathbb{R}^d$ are learnable vectors, and $\lambda_{\text{init}} \in (0, 1)$ is a constant for initializing λ . As demonstrated by the ablation study, setting λ_{init} to 0.8 provides the best overall performance.

The outputs from different heads are then concatenated and projected to form the final multi-head representation, which is expressed as

$$\mathbf{D}_i = \mathbf{A}_i - \lambda \hat{\mathbf{A}}_i, \quad (10)$$

$$\text{head}_i = \mathbf{D}_i \mathbf{V}_i, \quad (11)$$

$$\widehat{\text{head}}_i = (1 - \lambda_{\text{init}}) f_{\text{norm}}(\text{head}_i), \quad (12)$$

$$\text{MultiHead}(X_{\text{seq}}) = [\widehat{\text{head}}_1, \dots, \widehat{\text{head}}_h] \mathbf{W}^o, \quad (13)$$

where λ_{init} is the same constant as in Eq. (9), $\mathbf{W}^o \in \mathbb{R}^{hw \times hw}$ is a learnable projection matrix, and $[\cdot]$ denotes concatenation of the head outputs along the channel dimension. Here, each normalized head output is multiplied by the scaling factor $(1 - \lambda_{\text{init}})$ to balance gradient magnitudes and maintain training stability

$$F'_I = f_{\text{unflatten}}(\text{MultiHead}(X_{\text{seq}})). \quad (14)$$

Finally, the sequence features are reshaped back to two-dimensional feature maps with the original spatial dimensions through the $f_{\text{unflatten}}$ operation, yielding the output features of the MHDA module.

3) *Wavelet Cross Attention (WCA)*: To compensate for the limited capability of global conditional guidance in capturing fine-grained details, a WCA module is employed in each decoder stage. Although the skip connections in U-Net can propagate multiscale information from the encoder to the decoder, the transmitted features, which are mainly modulated by FiLM, primarily provide coarse structural guidance and are insufficient for recovering high-frequency details. Since wavelet decomposition can explicitly separate image components at different frequencies, WCA incorporates wavelet features into the decoder through cross-attention, providing fine-grained textural and structural guidance. The detailed architecture of WCA is illustrated in Fig. 2(c).

Specifically, the high-resolution MSI ($\mathcal{Y}$) and low-resolution HSI ($\mathcal{X}$) are separately subjected to a discrete wavelet transform using the DB1 wavelet basis. This process decomposes each image into four separate frequency components, i.e., a

low-low frequency approximation and three high-frequency details (low-high (LH), high-low (HL), and high-high (HH))

$$LL_{\mathcal{X}}, LH_{\mathcal{X}}, HL_{\mathcal{X}}, HH_{\mathcal{X}} = \text{DB1}(\mathcal{X}), \quad (15)$$

$$LL_{\mathcal{Y}}, LH_{\mathcal{Y}}, HL_{\mathcal{Y}}, HH_{\mathcal{Y}} = \text{DB1}(\mathcal{Y}). \quad (16)$$

Note that we preserve the low-frequency principal components of the HSIs and the high-frequency detail components of the MSIs (LH, HL, and HH) for injection into the feature input module as wavelet features, which can be expressed as follows

$$c = [LL_{\mathcal{X}}, LH_{\mathcal{Y}}, HL_{\mathcal{Y}}, HH_{\mathcal{Y}}], \quad (17)$$

where c denotes the conditional feature. To facilitate the interaction between wavelet features and backbone feature maps, a linear cross-attention mechanism is introduced. Specifically, the query matrix $\mathbf{Q}$ is obtained from the decoder features F_l through a linear projection, while the key matrix $\mathbf{K}$ and value matrix $\mathbf{V}$ are derived from the conditional features c via independent linear mappings

$$\mathbf{Q} = f_{\text{conv}}(F_l), \quad (18)$$

$$\mathbf{K}, \mathbf{V} = f_{\text{chunk}}(f_{\text{conv}}(c)). \quad (19)$$

Following the linear attention formulation, $\mathbf{K}$ is first normalized by the softmax function and multiplied with the transpose of $\mathbf{V}$ to construct the attention matrix $\mathbf{A}$. The query features $\mathbf{Q}$ are then weighted by $\mathbf{A}$ to produce the intermediate cross-attention representation

$$\mathbf{A} = \text{softmax}(\mathbf{K}) \otimes \mathbf{V}^T, \quad (20)$$

$$\mathbf{O} = \mathbf{A} \otimes \mathbf{Q}. \quad (21)$$

Finally, we enhance the cross-attention output $\mathbf{O}$ through a feedforward layer to derive the final output feature F'_l ,

$$F'_l = \text{FFN}(\mathbf{O}) + \mathbf{O}, \quad (22)$$

where the feedforward layer FFN consists of the convolution of 1×1 and ECA attention.

D. DDIM Fast Sampling

A key limitation of the DDPM is that the reverse denoising process requires an identical number of sampling steps as the forward diffusion process, which results in a time-intensive image generation procedure and substantial computational expense. To accelerate the image generation process, the proposed method employs DDIM [47] to accelerate the sampling process.

The core advantage of a DDIM lies in its non-Markovian formulation of the reverse process, which enables fast sampling. The single-step sampling process of the DDIM can be expressed as

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_{\theta}(x_t, t, c) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \epsilon_{\theta}(x_t, t, c) + \sigma_t^2 \epsilon, \quad (23)$$

where σ_t is the predefined noise standard deviation related to timestep t , $x_{\theta}(x_t, t, c)$ represents the model's estimate of the original data x_0 , and the noise term $\epsilon_{\theta}(x_t, t, c)$ is calculated as

$$\epsilon_{\theta}(x_t, t, c) = \frac{x_t - \sqrt{\bar{\alpha}_t} x_{\theta}(x_t, t, c)}{\sqrt{1 - \bar{\alpha}_t}}. \quad (24)$$

As formulated in Eq. (24), the denoised result x_{t-1} comprises three parts, i.e., the prediction of the original data $x_{\theta}(x_t, t, c)$, the noise estimate $\epsilon_{\theta}(x_t, t, c)$, which is calculated according to Eq. (25), and a randomly sampled standard Gaussian noise $\epsilon \sim \mathcal{N}(0, 1)$.

To achieve effective acceleration, the DDIM utilizes a sparse sampling timestep sequence $\tau = \{\tau_1, \tau_2, \dots, \tau_S\}$, where the number of DDIM sampling steps S is considerably less than the total number of steps T in the original diffusion process. Specifically, for any sampling step $\tau_i \in \tau$, the model predicts $x_{\theta}(x_{\tau_i}, \tau_i, c)$ as follows

$$x_{\theta}(x_{\tau_i}, \tau_i, c) := \frac{x_{\tau_i} - \sqrt{1 - \bar{\alpha}_{\tau_i}} \epsilon_{\theta}(x_{\tau_i}, \tau_i, c)}{\sqrt{\bar{\alpha}_{\tau_i}}}. \quad (25)$$

This sparse sampling strategy facilitates a substantial improvement in the model's sampling efficiency while maintaining the quality of the generated outputs.

III. EXPERIMENTS

A. Implementation Details

The proposed method is implemented in PyTorch and optimized using the AdamW optimizer. The training objective is to minimize the ℓ_1 loss between the predicted image $\hat{Z}$ and the target image Z . The model is trained for 1.5×10^5 iterations with a batch size of 32. The initial learning rate is set to 2×10^{-4} and decayed by a factor of 0.2 at 50k, 75k, and 100k iterations. All experiments are conducted on a Windows 11 platform equipped with a RTX 5070Ti GPU with 16 GB VRAM.

For the diffusion model, a cosine noise schedule is adopted with the total number of diffusion steps set to $T = 1000$, while DDIM sampling is performed using 25 steps. The backbone network follows a U-Net architecture with four encoder and four decoder stages. The base channel number is set to 32, and the channel multipliers for the encoder are configured as [1, 2, 2, 4], with the decoder adopting a symmetric configuration.

B. Datasets

To comprehensively evaluate the generalization ability of the proposed method, four publicly available datasets with different scene characteristics and scales are adopted, including CAVE [48],¹ Harvard [49],² Pavia Center [50],³ and Chikusei.⁴ Specifically, CAVE mainly contains indoor scenes, Harvard covers natural scenes, Pavia Center represents complex urban environments, and Chikusei is a large-scale airborne hyperspectral remote sensing dataset. These datasets provide diverse spectral characteristics and spatial structures, enabling a comprehensive evaluation of spectral preservation, spatial detail recovery, and cross-scene generalization. To ensure fair comparisons, the CAVE, Harvard, and Pavia Center datasets are

¹<https://cave.cs.columbia.edu/repository/Multispectral>

²<http://vision.seas.harvard.edu/hyperspec/index.html>

³https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University

⁴<https://naotoyokoya.com/Download.html>

TABLE II

EXPERIMENTAL DATASETS INFORMATION. (-/-BANDS) INDICATES THE NUMBER OF SPECTRAL BANDS FOR HSIS AND MSIS, RESPECTIVELY. VALUES IN PARENTHESES INDICATE THE SPATIAL RESOLUTION OF EACH SAMPLE

Dataset	Train	Valid	Test
CAVE(31/3 Bands)	3136 (64×64)	784 (64×64)	11 (512×512)
Harvard(31/3 Bands)	3136 (64×64)	784 (64×64)	10 (1000×1000)
Pavia Centre(92/4 Bands)	1284 (64×64)	142 (64×64)	4 (256×256)
Chikusei(128/8 Bands)	832 (64×64)	64 (64×64)	8 (256×256)

directly adopted from the publicly available benchmark dataset released by PSRT.⁵

The Chikusei dataset contains 128 spectral bands and covers various land-cover types, including farmland, buildings, roads, and vegetation. Following the Wald protocol, MSI is generated by spectral degradation using the WorldView-2 spectral response function, while LRHSI is obtained by applying a Gaussian blur with a kernel size of 7×7 and a standard deviation of 2, followed by spatial downsampling. The original HSI is used as the ground truth for quantitative and qualitative evaluation. For all datasets, the spatial downsampling factor is set to 4.

The detailed parameters and division schemes of each dataset are summarized in Table II.

C. Compared Methods and Evaluation Metrics

To evaluate the proposed method objectively, we performed extensive comparisons of the following mainstream fusion approaches. In addition to visual comparisons, various evaluation metrics were employed for comparative analysis, including Peak Signal-to-Noise Ratio(PSNR) [65], Spectral Angle Mapper (SAM) [66], Erreur Relative Globale Adimensionnelle de Synthèse(ERGAS) [67], Structural Similarity Index Measure(SSIM) [59], $Q2^n$ [68], Correlation Coefficient(CC) [69], Spatial Correlation Coefficient(SCC) [70], and Root Mean Squared Error(RMSE) [20]. We also performed visual comparisons to complement the quantitative analysis.

Comparison methods encompass both traditional model-based algorithms and data-driven deep learning algorithms. To ensure a fair comparison, all deep learning models were trained and evaluated using the same dataset and experimental configuration. Table III shows the model complexity and efficiency data, including the number of parameters, FLOPs, and runtime.

D. Experimental Results on the CAVE Dataset

As shown in Fig. 3, the compared methods exhibit noticeable differences in spatial detail restoration and spectral fidelity. Conventional methods suffer from obvious distortions in edge regions and local textures, particularly in the red halo area. Although deep learning-based methods generally provide better spatial reconstruction, some methods still exhibit artifacts or color shifts. For instance, PSRT, DDIF, and SMGU show pronounced responses around the halo boundaries in the residual maps, indicating insufficient modeling

⁵<https://github.com/shangqideng/PSRT>

TABLE III

COMPARISON OF MODEL COMPLEXITY AND RUNTIME AMONG DIFFERENT METHODS. ALL RESULTS ARE OBTAINED ON THE CAVE DATASET, AND THE REPORTED FLOPS AND INFERENCE TIMES ARE BASED ON 64×64 IMAGE PATCHES

Method	Params (M)	FLOPs (G)	Inference Time (s)	Training Time (10^3 s)
GSA [51] ²⁰⁰⁷	/	/	0.0107	/
CNMF [52] ²⁰¹¹	/	/	0.3012	/
FUSE [53] ²⁰¹⁵	/	/	0.0237	/
U2Net [54] ²⁰²³	0.7850	3.0911	0.0075	41.007
PSRT [18] ²⁰²³	0.2475	1.1385	0.0128	44.011
DDIF [55] ²⁰²⁴	6.1237	68.7390	1.3271	30.674
MIMOSST [22] ²⁰²⁴	4.9830	1.5414	0.0098	11.519
DCT [56] ²⁰²⁴	8.1454	81.7500	0.0927	167.720
MIMFormer [57] ²⁰²⁴	16.2426	6.0726	0.0094	14.333
SMGU [58] ²⁰²⁵	0.7594	5.8365	0.0092	18.188
OTPNNet [59] ²⁰²⁵	2.3962	9.8149	0.0077	14.268
AMSF [60] ²⁰²⁵	7.1611	13.2756	0.0329	39.582
ASMNet [61] ²⁰²⁵	1.2484	0.9508	0.0030	5.485
SSDT [62] ²⁰²⁵	5.9634	4.5872	0.0236	26.394
DPFormer [63] ²⁰²⁶	1.4646	2.5606	0.0240	23.078
CLSNet [64] ²⁰²⁶	4.0621	3.9870	0.0345	41.882
Ours	6.1988	70.9514	1.5033	31.596

of spectral characteristics in high-intensity regions. In contrast, MIMOSST, CLSNet, and the proposed method achieve more favorable visual results. Moreover, the proposed method exhibits weaker responses in the residual maps, suggesting a reconstruction closer to the ground truth.

The band-wise evaluations in Fig. 4 further support these observations. The proposed method achieves the highest spectral consistency with the reference image and consistently competitive per-band PSNR and UIQI values, especially in the middle and high spectral bands.

More visual comparisons on the CAVE dataset are provided in Appendix C of the Supplementary Material.

E. Experimental Results on the Harvard Dataset

As shown in Fig. 5, deep learning-based methods generally provide more stable results than conventional methods. Among them, the proposed method restores complex textures and edge structures more faithfully and exhibits weaker residual responses, indicating a reconstruction closer to the ground truth.

The band-wise results in Fig. 6 further confirm these observations, showing that the proposed method maintains high spectral consistency with the reference image and competitive per-band PSNR and UIQI values.

Table IV shows that the proposed method achieves the best or tied-best results in PSNR, SSIM, CC, and RMSE and delivers competitive performance in the remaining metrics, demonstrating a favorable balance between spatial detail restoration and spectral fidelity. Nevertheless, the proposed method is slightly inferior to MIMOSST in SAM and ERGAS. This performance gap may be attributed to the larger image size and richer texture information of the Harvard dataset, which increase the difficulty of preserving spectral consistency in complex scenes.

F. Experimental Results on the Pavia Center Dataset

As shown in Fig. 7, conventional methods suffer from noticeable distortions in edge regions and texture-rich areas,

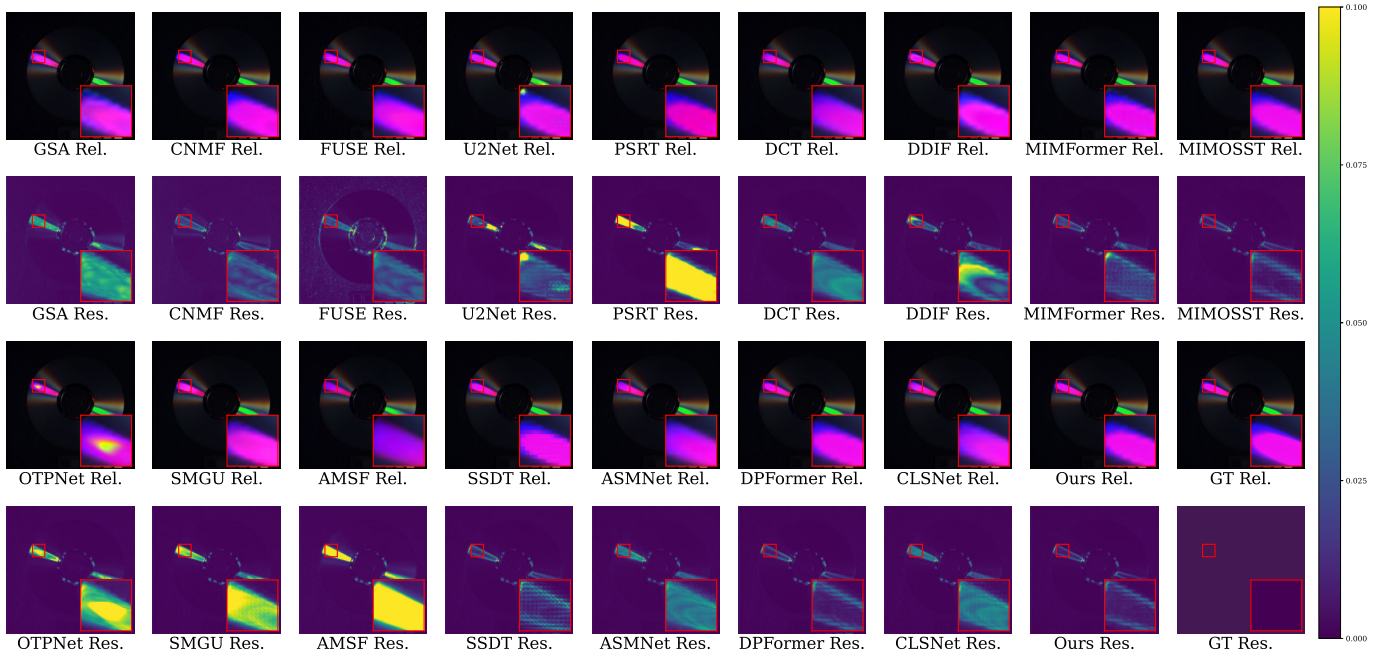


Fig. 3. Comparison of fusion results produced by different methods on the CAVE dataset (cd_ms). * Rel indicates a false-color composite generated using bands R-25, G-15, and B-5. * Res denotes the mean residual map, obtained by subtracting the fused image from the reference image and averaging across the spectral dimension.

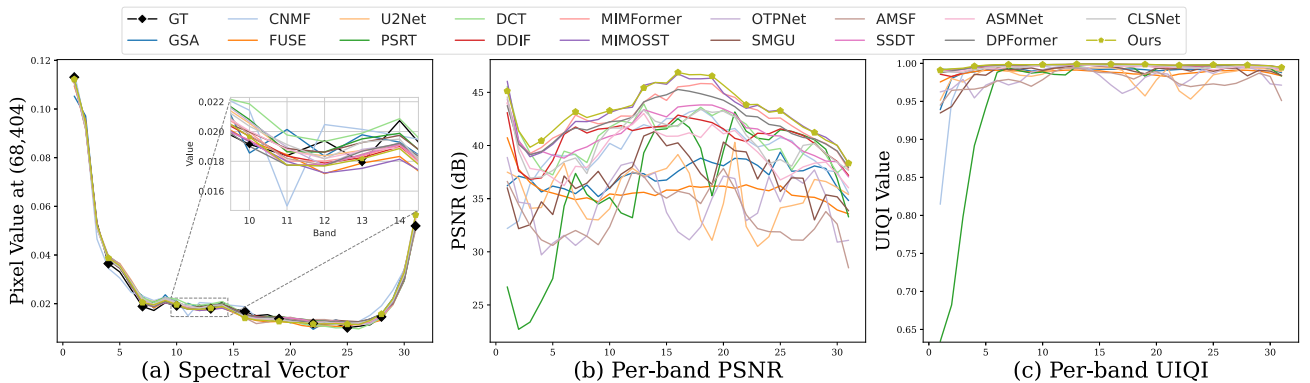


Fig. 4. Detailed evaluation of fusion performance. (a) Spectral vectors at pixel (163, 209) from different fused images. (b) Band-wise PSNR of different fused images. (c) Band-wise UIQI of different fused images.

whereas deep learning-based methods generally provide more stable fusion results. Nevertheless, some methods, such as MIMFormer and AMSF, still exhibit pronounced residual responses in local textured regions, while U2Net, PSRT, and DDIF show insufficient recovery of fine edge details. In contrast, the proposed method and SMGU achieve more favorable visual results, with the proposed method exhibiting weaker residual responses and more faithful reconstruction of fine-grained structures.

The band-wise results in Fig. 8 further confirm these observations, showing that the proposed method maintains high spectral consistency with the reference image and achieves competitive per-band PSNR and UIQI values.

Table IV summarizes the average quantitative results on the Pavia Center dataset. The proposed method achieves the best or tied-best performance in PSNR, SSIM, SAM, ERGAS, CC, SCC, and RMSE, while only slightly trailing OTPNet

in Q2N. These results demonstrate the effectiveness of the proposed method in recovering high-frequency spatial details while preserving spectral fidelity.

G. Experimental Results on the Chikusei Dataset

As shown in Fig. 9, the Chikusei dataset contains large areas of farmland and elongated structures, posing challenges for both weak-texture reconstruction and edge preservation. Conventional methods suffer from noticeable spectral distortions and structural errors, whereas deep learning-based methods generally provide more stable fusion results. Among them, methods such as MIMFormer, MIMOSST, SMGU, and CLSNet achieve favorable visual quality. In contrast, the proposed method exhibits weaker residual responses and more faithfully restores farmland boundaries and elongated structures, indicating a reconstruction closer to the ground truth.

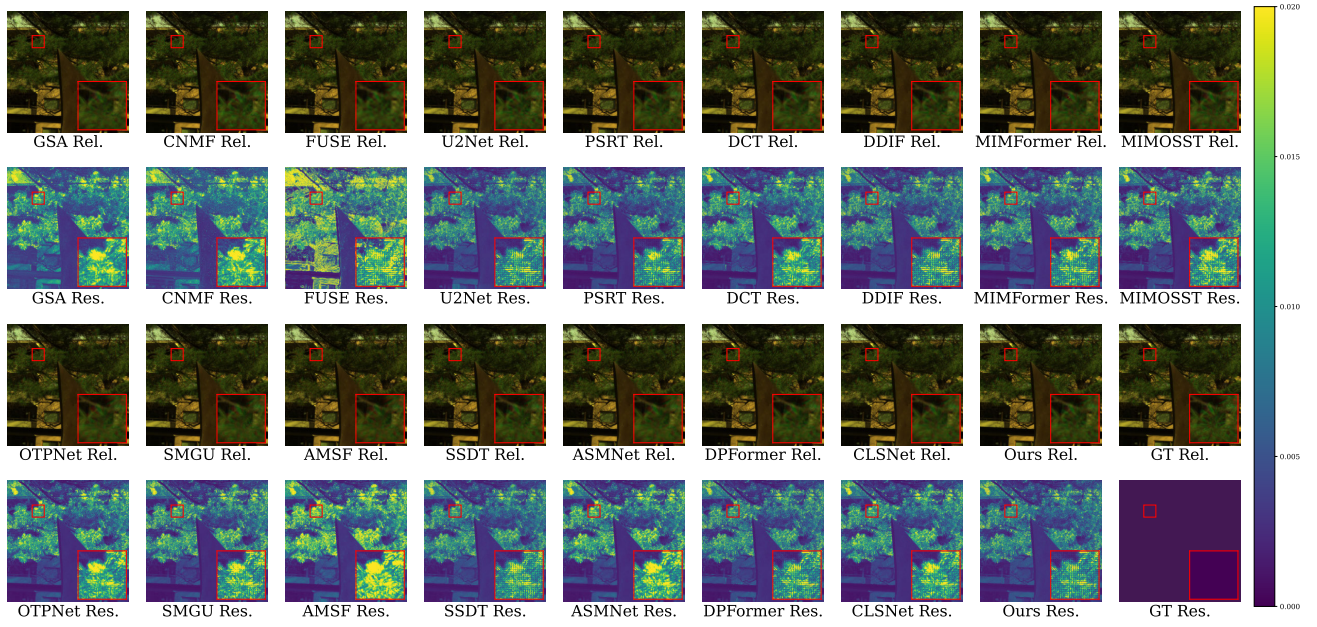


Fig. 5. Comparison of fusion results produced by different methods on the Harvard dataset (CZ_hdsdb\ imgb7). * Rel indicates a false-color composite generated using bands R-26, G-16, and B-6. * Res denotes the mean residual map, obtained by subtracting the fused image from the reference image and averaging across the spectral dimension.

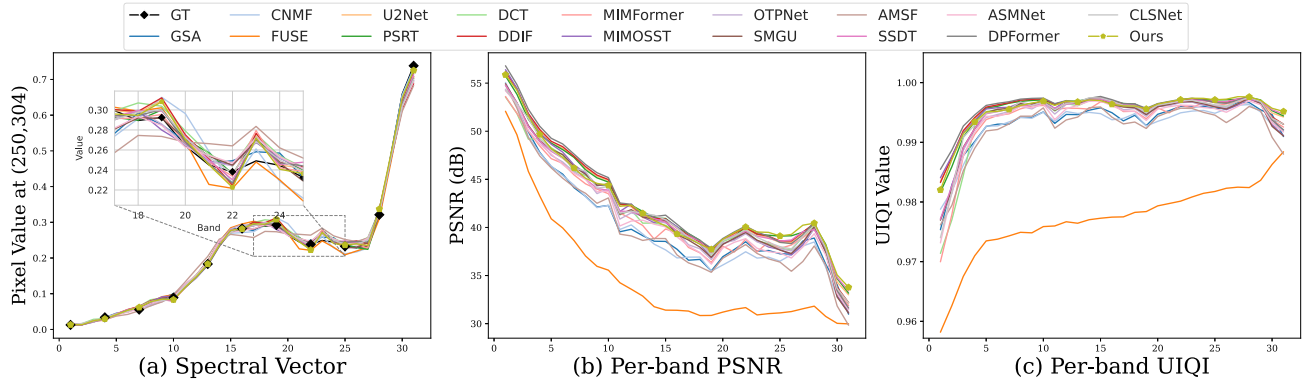


Fig. 6. Detailed evaluation of fusion performance. (a) Spectral vectors at pixel (68, 404) from different fused images. (b) Band-wise PSNR of different fused images. (c) Band-wise UIQI of different fused images.

Table IV summarizes the average quantitative results on the Chikusei dataset. The proposed method achieves the best performance across all evaluation metrics. Compared with strong competitors such as CLSNet, SMGU, and MIMOSST, the proposed method consistently delivers higher reconstruction accuracy and better structural and spectral preservation, demonstrating its superior overall fusion performance on large-scale remote sensing scenes.

H. Ablation Study

An ablation study was performed to evaluate the contributions of the three key modules, i.e., the FiLM, WCA, and MHDA modules, implemented in the proposed model, as well as the overall performance gains achieved through the residual learning strategy. In addition, an experimental analysis was performed to investigate the impact on model performance of varying the MHDA module's parameters. Furthermore,

the influence of different training objectives for the diffusion model on the fusion results was also examined.

1) *Module Ablation Experiment and Analysis*: The proposed method employs two conditional feature modulation modules, i.e., the FiLM and WCA modules. The FiLM module is mainly used for coarse-grained conditional feature injection during the encoder phase, and the WCA module is designed to refine high-frequency details during the decoder phase. For comparison, in the absence of the FiLM and WCA modules, a conventional baseline approach concatenates the conditional information with the model's primary input along the channel dimension. In addition, to enhance the efficiency of the feature extraction process in the attention mechanism, the MHDA module is introduced as an improvement over the conventional attention. Specifically, four configurations were considered in this evaluation.

1) Concatenating the conditional information with the primary input along the channel dimension

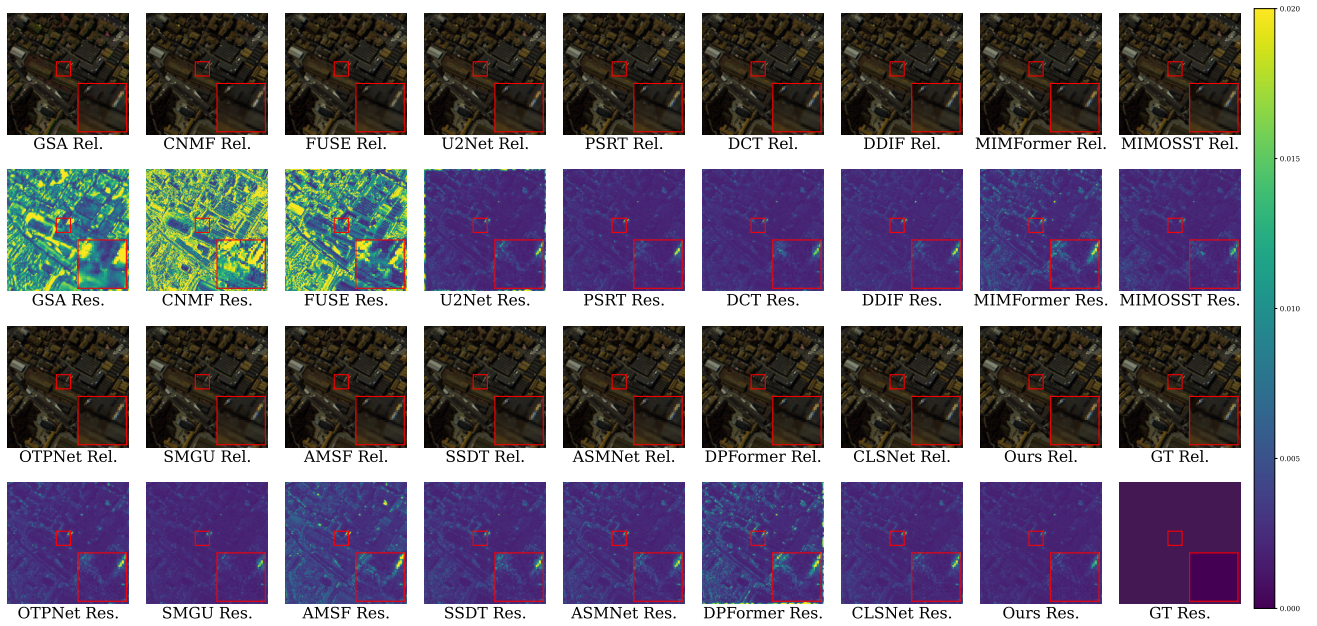


Fig. 7. Comparison of fusion results produced by different methods on the Pavia Center dataset. * Rel indicates a false-color composite generated using bands R-57, G-34, and B-3. * Res denotes the mean residual map, obtained by subtracting the fused image from the reference image and averaging across the spectral dimension.

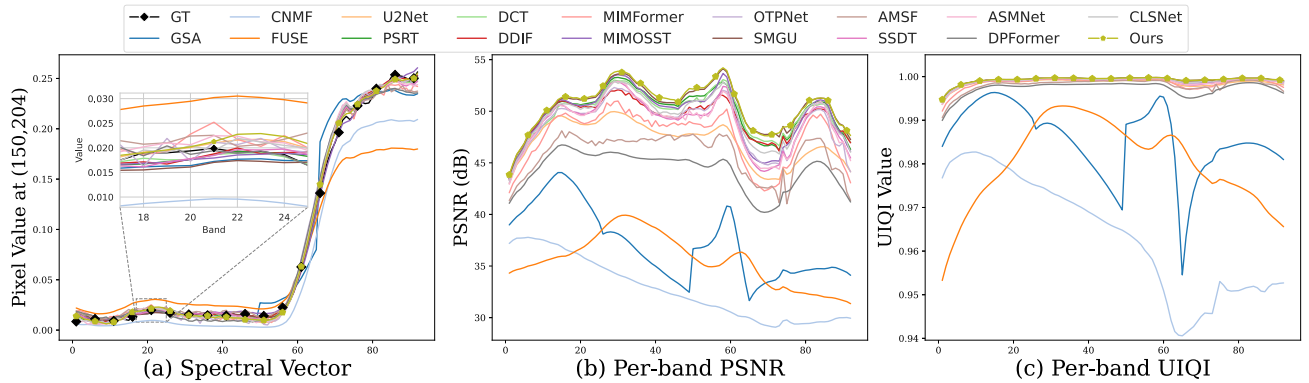


Fig. 8. Detailed evaluation of fusion performance. (a) Spectral vectors at pixel (68, 204) from different fused images. (b) Band-wise PSNR of different fused images. (c) Band-wise UIQI of different fused images.

TABLE V
ABLATION STUDY OF FiLM, WCA, AND MHDA MODULES
ON FUSION PERFORMANCE

FiLM	WCA	MHDA	PSNR $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$
$\times$	$\times$	$\times$	47.1131	3.9675	1.9844
$\checkmark$	$\times$	$\times$	47.6964	3.3062	1.8308
$\checkmark$	$\checkmark$	$\times$	48.6462	2.8819	1.5692
$\checkmark$	$\checkmark$	$\checkmark$	48.9966	2.6603	1.2689

TABLE VI
ABLATION STUDY OF RESIDUAL LEARNING ON FUSION PERFORMANCE

Residual Learning	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2N $\uparrow$	RMSE $\downarrow$
$\times$	42.2368	0.9903	4.8480	2.8232	0.9035	2.2105
$\checkmark$	48.9966	0.9961	2.6603	1.2689	0.9384	0.9603

without incorporating the FiLM, WCA, or MHDA modules.

- Integrating the FiLM module into the UNet architecture for conditional injection.
- Building upon (2) by further integrating the WCA module.
- Incorporating all three FiLM, WCA, and MHDA modules.

To ensure a fair comparison, each configuration was trained independently until convergence, and the performance was evaluated on the CAVE dataset using the PSNR, SAM, and ERGAS metrics. As shown in Table V, the results demonstrate a consistent improvement in the performance metrics with the progressive inclusion of the FiLM, WCA, and MHDA modules, thereby demonstrating their complementary benefits.

2) *Residual Learning*: The quantitative results presented in Table VI clearly demonstrate the positive impact of residual

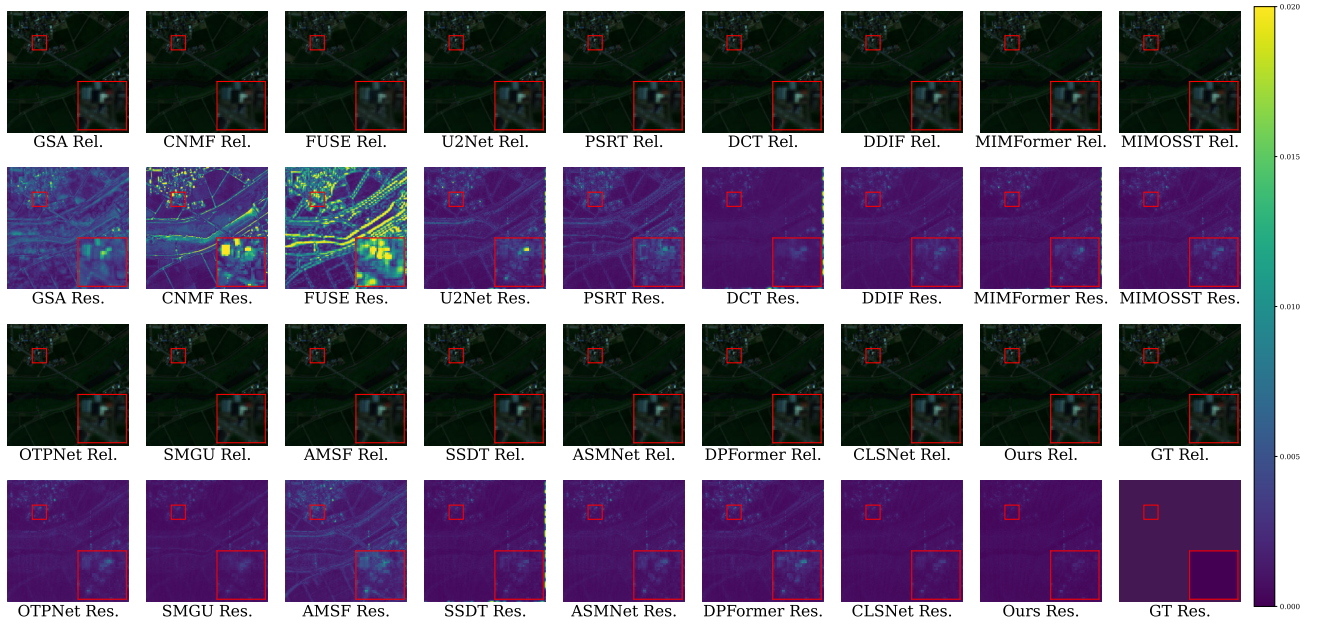


Fig. 9. Comparison of fusion results produced by different methods on the Chikusei dataset. * Rel indicates a false-color composite generated using bands R-60, G-40, and B-20. * Res denotes the mean residual map, obtained by subtracting the fused image from the reference image and averaging across the spectral dimension.

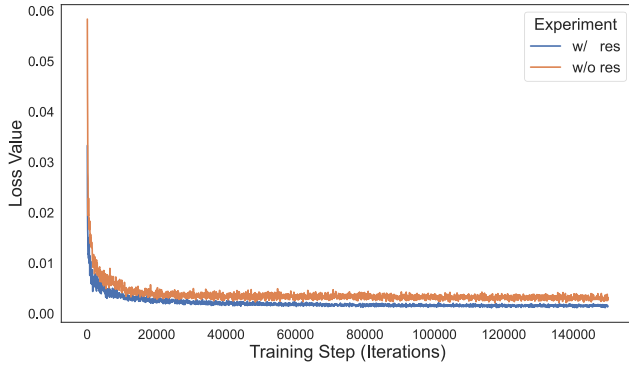


Fig. 10. The change in loss on the CAVE dataset over iterations, with and without the residual learning technique (w/ res, w/o res).

learning on the performance of the model. Compared with the baseline model, which directly predicts the target image, the residual learning model achieved superior performance across all evaluation metrics. In addition, the training loss curves shown in Fig. 10 corroborates this finding, i.e., the residual learning model converged faster and obtained a lower final loss value.

The underlying reason for this performance gain can be attributed to the differences in the complexity of the data distribution between the residual images and complete target images. As reported in related literature [55], information entropy is an effective indicator to quantify the complexity of image data distributions. Our calculations reveal that residual images possess markedly lower information entropy than their corresponding complete high-resolution HSIs, which suggests that generating residuals constitutes a simpler learning task, enabling the neural network to fit and optimize more efficiently, thereby leading to enhanced performance.

3) λ_{init} *Parameter Analysis*: The parameter λ_{init} is a key hyperparameter in the differential attention mechanism, determines the initial weight λ of the differential operation within the module. Two main strategies are employed for setting λ_{init} :

- 1) **Constant initialization**: When instantiating the module, λ_{init} is preset to a fixed constant value.
- 2) **Dynamic depth-aware initialization**: The λ_{init} value is calculated dynamically based on the depth of the network layer in which it resides. Here, the specific formula is given in Eq. (26), where e and l denote the base of the natural logarithm and the index of the network layer's depth, respectively.

$$\lambda_{init} = 0.8 \times 0.6e^{-0.3l}. \quad (26)$$

We performed an ablation study to investigate the effect of λ_{init} on the performance of the final model. A set of discrete fixed values [0.2, 0.4, 0.6, 0.8] was selected and compared against the dynamic depth-aware initialization strategy. To shorten the experimental period, all models were trained for 10×10^3 iterations. The results indicate that λ_{init} has a substantial impact on model performance. As shown in Fig. 11, among the configurations considered in this experiment, setting λ_{init} to a fixed value of 0.8 yielded the best performance across all metrics.

4) *Training Objective Analysis*: The training process of diffusion models can be formulated with three distinct prediction targets, i.e., noise prediction (ϵ -prediction), original data prediction (x_0 -prediction), and velocity prediction (v -prediction) [32]. Note that these strategies differ in terms of their emphasis on model optimization and performance.

To systematically assess the impact of each training objective on the hyperspectral image fusion task, we trained a modified version of our model independently for each

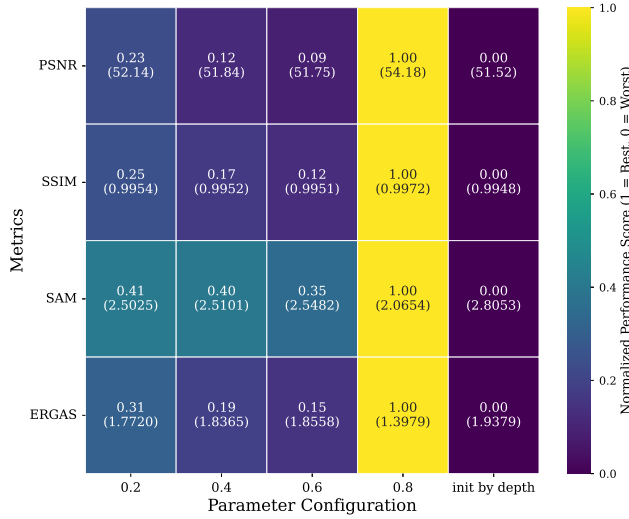


Fig. 11. Normalized heatmaps of fusion performance metrics. These experiments were conducted on the CAVE dataset, where lambda init values were set to [0.2, 0.4, 0.6, 0.8] and dynamic initialization based on network depth. To reduce computational cost, all networks shown in the figures were trained for 10×10^3 iterations.

TABLE VII

IMPACT OF TRAINING OBJECTIVES IN DIFFUSION MODELS ON TARGET IMAGE FUSION PERFORMANCE

Training objects	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2N $\uparrow$	RMSE $\downarrow$
ϵ	47.3262	0.9803	4.9739	1.7590	0.9025	1.4700
v	47.4428	0.9820	4.6980	1.6753	0.9170	1.2251
x_0	48.9966	0.9961	2.6603	1.2689	0.9384	0.9603

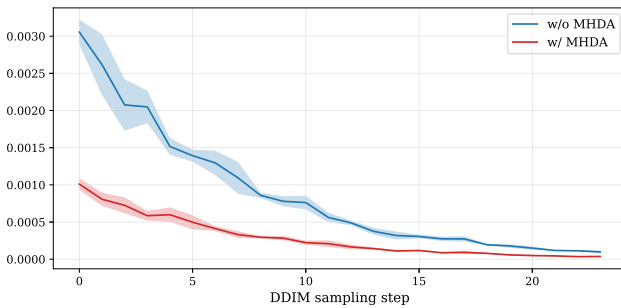


Fig. 12. Effect of MHDA on generation stability. Curves show MSE with (red) and without (blue) MHDA over DDIM sampling steps; shaded areas indicate standard deviation. MHDA consistently lowers errors and fluctuations, mitigating hallucinated content.

objective until it converged. As shown in Table VII, this model obtained the best performance across all evaluation metrics when the objective was set to predict the target image, which suggests that a diffusion model performs better for this specific task when it is guided to directly learn and generate the final image rather than predicting intermediate variables.

5) *MHDA for Suppressing Generated Hallucinations*: To further investigate the effect of MHDA on diffusion-induced hallucinations, we evaluate the stability of the reverse sampling process by measuring the mean squared error and standard deviation between the predicted and target images during

TABLE VIII

COMPARISON OF RECONSTRUCTION PERFORMANCE AND INFERENCE TIME BETWEEN DDPM AND DDIM SAMPLING, WHERE NFE DENOTES THE NUMBER OF FUNCTION EVALUATIONS (SAMPLING STEPS)

Sampler	NFE	Time(s) $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$
DDPM	1000	276.4191	48.9966	0.9962	2.6600	1.2689
DDIM	25	6.7212	48.9966	0.9961	2.6603	1.2689

TABLE IX

EFFECT OF DIFFERENT NOISE COEFFICIENTS AND RANDOM SEEDS ON THE GENERATION PROCESS

σ_t	PSNR		SAM	
	Mean	Std	Mean	Std
0.00	48.99664	0.00005	2.66035	0.00002
0.25	48.99662	0.00013	2.66038	0.00011
0.50	48.99636	0.00052	2.66051	0.00036
0.75	48.99662	0.00017	2.66037	0.00039
1.00	48.99660	0.00011	2.66048	0.00021

sampling. As shown in Fig. 12, MHDA consistently achieves lower errors and smaller fluctuations, indicating more stable convergence and reduced error accumulation during iterative denoising.

To gain further insight, Fig. 13 visualizes the attention maps of the two attention branches in MHDA and their differential outputs. Compared with the original attention maps, the differential attention is considerably sparser and mainly focuses on structural boundaries and salient regions, while common responses and noisy activations are effectively suppressed. This suggests that MHDA preserves discriminative spectral-spatial interactions while reducing redundant feature aggregation.

Overall, MHDA suppresses attentional noise and alleviates the accumulation of spurious features during iterative denoising, thereby improving generation stability and reducing hallucinated structures.

6) *Effectiveness of DDIM Fast Sampling*: Table VIII compares the reconstruction performance and inference time of DDPM and DDIM sampling. With only 25 sampling steps, DDIM achieves nearly identical performance to DDPM (1000 steps) across all metrics, while reducing the inference time from 276.42 s to 6.72 s. This demonstrates that DDIM substantially improves sampling efficiency without sacrificing fusion quality.

7) *Effect of Random Seeds and DDIM Stochasticity on Generation Results*: To investigate the impact of randomness on the proposed framework, we further evaluate the reconstruction performance under different DDIM noise coefficients and random seeds. In DDIM sampling, the parameter σ_t in Eq. (23) controls the amount of stochastic noise injected at each reverse step, where $\sigma_t = 0$ corresponds to deterministic sampling and larger values introduce stronger randomness. Specifically, σ_t is varied over $\{0, 0.25, 0.5, 0.75, 1\}$ and seed is chosen from $\{0, 42, 123, 456, 789\}$. The results are summarized in Table IX. As shown, the proposed method exhibits highly consistent performance across all settings, with only negligible variations

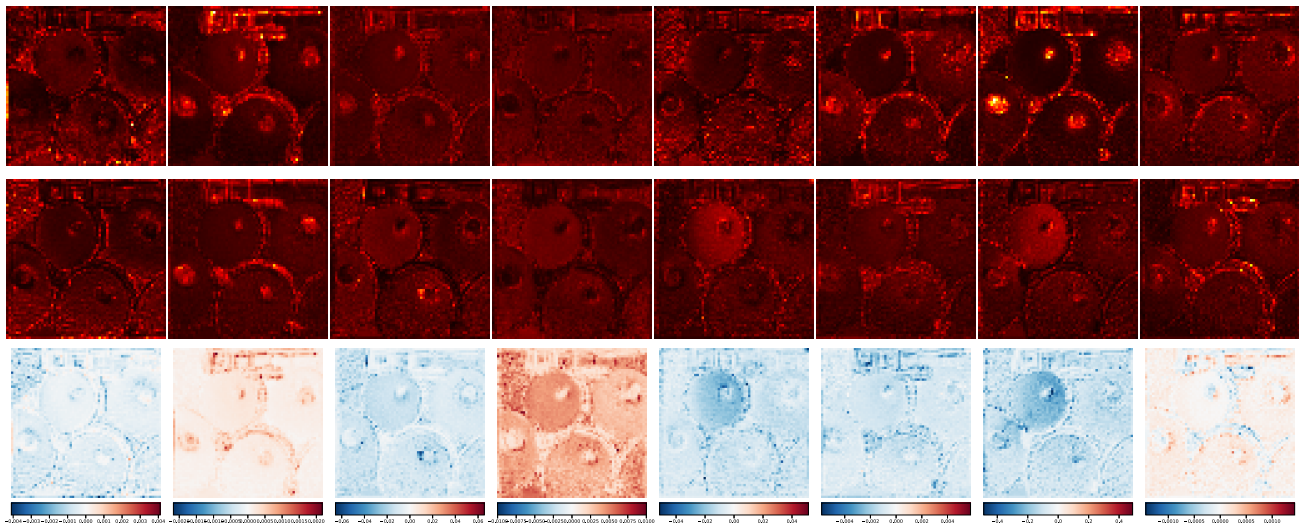


Fig. 13. Visualization of differential attention maps. The first two rows show the attention maps from two branches, and the third row shows the resulting differential attention maps. Each column corresponds to a different attention head.

in PSNR and SAM, indicating that the randomness introduced by DDIM sampling has little effect on fusion quality. This robustness can be attributed to the strong conditional guidance and the largely deterministic sampling process of DDIM, which effectively constrain the reverse process and make the reconstruction insensitive to different random initializations. As a result, the proposed framework maintains stable fusion performance under different random seeds and σ_t values.

I. Downstream Task Extension

To evaluate practical effectiveness, we conducted spectral unmixing as a downstream task. Detailed experimental setups, quantitative results, and visual comparisons (abundance maps and endmember curves) for the CAVE and Pavia Center datasets are provided in Appendix D of the Supplementary Material. These results consistently demonstrate the superior spectral fidelity of our method over state-of-the-art (SOTA) competitors.

IV. CONCLUSION

This paper presents a two-stage conditionally guided diffusion model for hyperspectral and multispectral image fusion. Equipped with a differential attention mechanism and a coarse-to-fine guidance strategy, the proposed method enhances fine-detail reconstruction while preserving spectral, spatial, and structural fidelity. Experimental results on multiple benchmark datasets demonstrate its superiority over existing state-of-the-art methods.

Despite its effectiveness, two limitations remain. First, the iterative sampling process of diffusion models leads to high computational cost and slow inference, hindering deployment in real-time and resource-limited scenarios. Second, the model is tailored to fixed spatial scales and spectral configurations, limiting its adaptability and generalization to diverse remote sensing data. Future work will investigate accelerated sampling techniques and a unified framework that supports arbitrary spatial scales and spectral bands, further improving

the flexibility and generalization capability of hyperspectral and multispectral image fusion models.

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